ME5413
Final Project Instructions

20 Feb 2026

Syllabus

6	20-Feb-26	Localisation (II) - 2D LiDAR based Back-End Optimisation and Mapping	Lecture	Final Project Released on 20 Feb 2026
Recess	27-Feb-26	Recess week	No Class	
7	6-Mar-26	Localisation (III) - 3D LiDAR/Visual/Multi-sensor Fusion SLAM	Lecture followed by tutorial	Tutorial (SLAM)
8	13-Mar-26	Planning (I) - Map Representations + Graph Search Methods	Quiz 2 followed by Lecture	**Quiz 2 –20% (SLAM)- 13 Mar 2026**
9	20-Mar-26	Planning (II) - Sampling Methods + Motion Primitives	Lecture	
10	27-Mar-26	Planning (III) - Optimization Methods + Challenges	Lecture, followed by tutorial	Tutorial (Planning)
11	3-Apr-26	Good Friday	No Class	
12	10-Apr-26	Quiz 3 followed by Hands-On Session AV Stack	Quiz 3 followed by Hands-On	**Quiz 3 - 20%- (Planning) - 10 April 2026**
13	17-Apr-26	Final Project Presentations	Student Presentations	**Final Project due - 12 Apr 2026 (Sun)** **Time: 2-9 pm (17 Apr 2026)** **Venue: Control & Mechatronics Lab**

- 20 Feb : Release Final project
- 27 Feb : Recess Week
- 6 Mar : SLAM Tutorial & Final Project Discussion

Final Project

Overview

In this simulated environment:

- 1 Jackal Robot
- X randomly generated boxes and cones
- 1 timed blockade
- 2 floors for exploration
- Obstacles to avoid
- 4 possible destinations

Your Task:

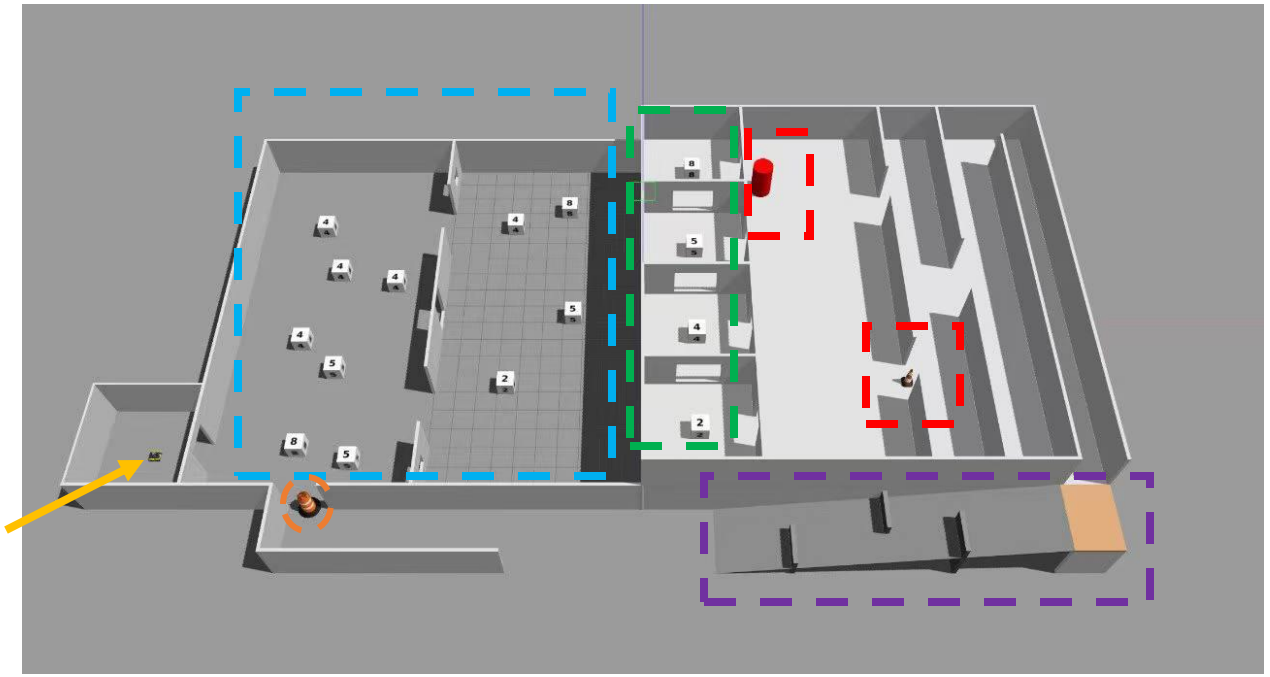
- Design a robot software stack that can:
  - Map the environment
  - Navigate autonomously
  - Perform the tasks on the right

1. Move & Count the number of Boxes

4. Avoid Obstacles

5. Find the Box with the least number of occurrences

"Jackal"



2. Publish & Unlock the 10s timed Blockade from the cone

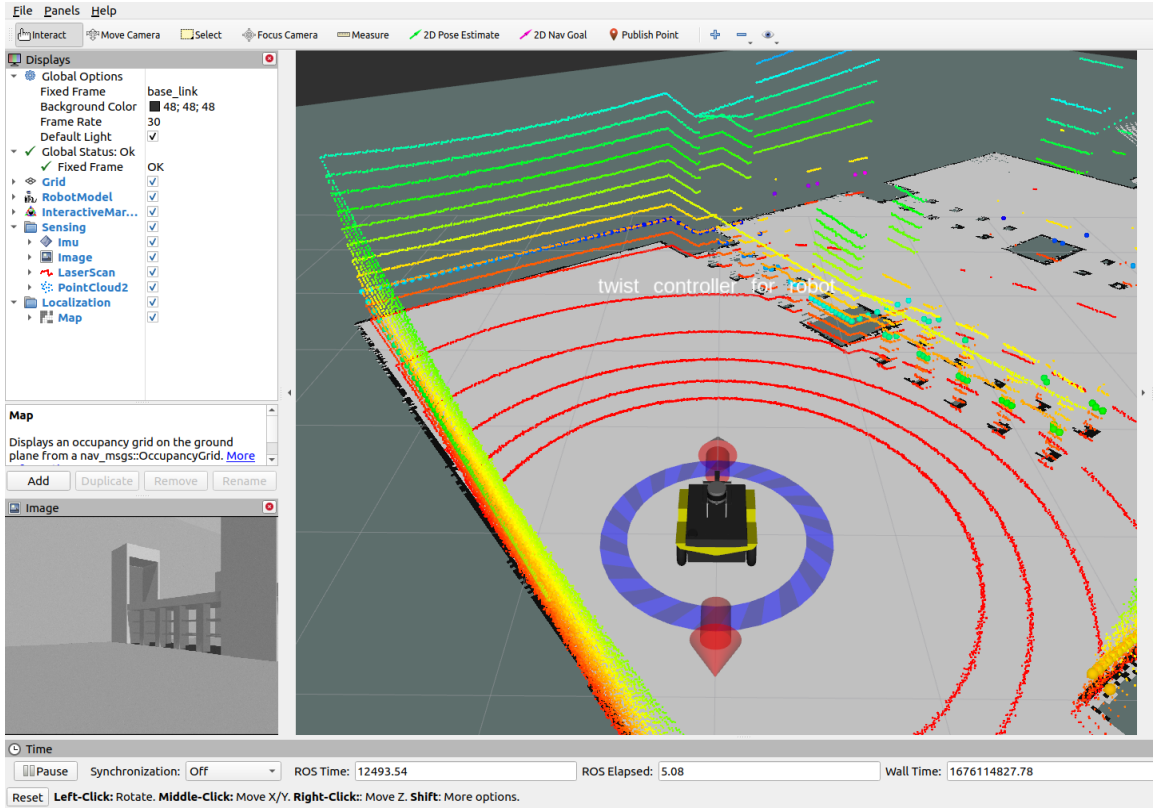
3. Travel to the second floor

Final Project

Tasks 1: Mapping

- Map the environment using any algorithm you like
- Evaluate the performance of your SLAM algorithm by comparing your estimated odometry with the ground truth odometry

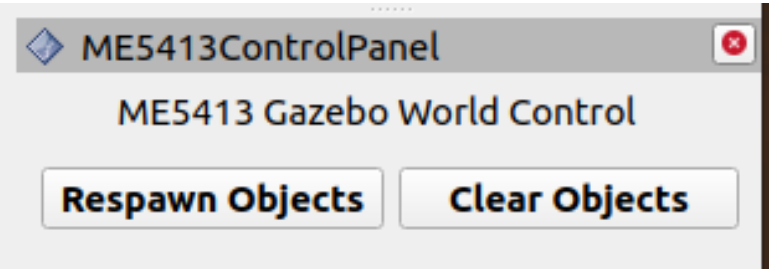
- In your report:
  - Describe your mapping pipeline in detail
  - Qualitatively and quantitatively analyse your SLAM performance (Figures and Tables)
  - Discuss the challenges you faced and your proposed solutions (with examples and comparisons)



Final Project

Tasks 2: Navigation

- Navigate your robot and perform the given sequence of tasks
- The score is calculated based on the number of tasks your robot can perform.
- In your report:
  - Describe your navigation pipeline in detail
  - Describe how your robot is designed to perform each task in detail
  - Qualitatively and quantitatively analyse the performance of your navigation stack in multiple metrics
  - Discuss the challenges you faced and your proposed solutions (with examples and comparisons)



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Submission

In your submission (GroupNumber.zip)

1. Report (.pdf, 10 pages max, appendices no limit)
2. Map file (any format)
3. A video showing your robot running along the designated route (.mp4, less than 50 Mb)
4. Presentation Slides (.pptx, less than 200 Mb)
5. Link to your Github repo with Well-documented commits

- Marking Rubric
- Peer Review

Rubric

Criteria	Ratings		Pts
Result Accuracy compared to the Ground Truth	10 Pts Full marks	0 Pts No marks	10 pts
Technical The correctness of your method	30 Pts Full marks	0 Pts No marks	30 pts
Effort The amount of work done	20 Pts Full marks	0 Pts No marks	20 pts
Code Style Readability; structure; naming convention; efficiency	20 Pts Full marks	0 Pts No marks	20 pts
Writing Clarity; comprehensiveness; conciseness	20 Pts Full marks	0 Pts No marks	20 pts
			Total points: 100

Final Project

Grouping

- Final Project groups:
  - Max 4 people per groups
  - Total – 14 groups
  - Unassigned groups will be assigned by the TAs by Friday, 20th Feb 9pm
- **Deadline: Sunday, 12 April 2026 23:59**
- Presentations:
  - Friday 17 April 2026

Groups (20)

Final Project Groups 1	5 / 6 students	
Final Project Groups 2	3 / 6 students	
Final Project Groups 3	Full 6 / 6 students	
Final Project Groups 4	Full 6 / 6 students	
Final Project Groups 5	5 / 6 students	
Final Project Groups 6	Full 6 / 6 students	

Final Project

Presentation

- Presentation: 10 mins [***based on submitted slides***]
- Live Demo: 5 mins
- Q&A: 3 mins

- Brief explanation of your robot system
  - Diagrams would be useful
  - The algorithms you used for each task
- Problems & Solutions
  - Describe the challenges you encountered and how you overcome them
  - Potential future work: how your system can be improved further
- Videos of your robot
  - Show your Demo!

Slot	Time	GROUP
		e.g Group 1
1	2:00 PM	
2	2:30 PM	
3	3:00 PM	
4	3:30 PM	
5	4:00 PM	
6	4:30 PM	
7	5:00 PM	
8	5:30 PM	
9	6:00 PM	
10	6:30 PM	
11	7:00 PM	
12	7:30 PM	
13	8:00 PM	
14	8:30 PM	
15	9:00 PM	

Final Project

FAQs

Q: Can I change the robot description file given in the project?

- Yes, you can modify the sensor configuration, by adding more sensors or new types of sensors, as well as their locations.
- However, you are not allowed to modify the mobile base.

Q: Can I use a different robot to do the mapping?

- No, it must be using the same robot mobile base. However, you can use a separate sensor configuration for mapping.